

Subscription Cohort Analysis

Power BI Project — Reference Module

This document covers every step taken in this project: data cleaning in Python, DAX measures created in Power BI, visuals built, and key decisions made along the way. Use it as a personal reference guide.

Dataset Overview

Table	Name	Rows	Columns	Purpose
df1	Subscription Cohort Dictionary	6	2	Data dictionary / field reference
df2	Subscription Cohort Data	3,069	6	Main analysis dataset

1. Python Data Cleaning (Jupyter Notebook)

1.1 — Handling Null Values in canceled_date

The canceled_date column had 1,065 null values representing active (non-churned) customers. Rather than filling with a string, we created a separate status column and kept nulls intact for date calculations.

```
# Create status label column
df2['status'] = df2['canceled_date'].apply(
    lambda x: 'Cancelled' if pd.isna(x) else 'Active'
)

# Create boolean churn flag
df2['is_churned'] = df2['canceled_date'].isna()
```

Why not fillna('Not Cancelled')? Because that converts a date column to a string object, breaking all date-based calculations downstream.

1.2 — Converting Date Columns to datetime

Both date columns were stored as object (string) dtype and needed conversion to datetime64.

```
df2['created_date'] = pd.to_datetime(df2['created_date'])
df2['canceled_date'] = pd.to_datetime(df2['canceled_date'])
```

1.3 — Creating Derived Columns

```
# Subscription duration in days (null for active customers)
df2['subscription_duration_days'] = (
    df2['canceled_date'] - df2['created_date']
).dt.days

# Cohort month — groups customers by their signup month
df2['cohort_month'] = df2['created_date'].dt.to_period('M')
```

cohort_month groups customers by the month they first subscribed, enabling cohort-level retention analysis.

1.4 — Final DataFrame Info

#	Column	Non-Null Count	Dtype	Notes
0	customer_id	3,069	int64	Primary key
1	created_date	3,069	datetime64[ns]	Converted from object
2	canceled_date	2,004	datetime64[ns]	Null = still active
3	subscription_cost	3,069	int64	Revenue basis
4	subscription_interval	3,069	object	Only value: "month"
5	was_subscription_paid	3,069	object	Paid/unpaid flag
6	subscription_duration_days	2,004	float64	Derived — churned only

7	is_churned	3,069	bool	Derived — churn flag
8	status	3,069	object	Derived — Active/Cancelled
9	cohort_month	3,069	period[M]	Derived — signup cohort

1.5 — Exporting Cleaned Data

```
# Convert period type to string before export
df2['cohort_month'] = df2['cohort_month'].astype(str)

df2.to_csv('cleaned_subscriptions.csv', index=False)
```

cohort_month must be cast to string before CSV export — Power BI cannot read the Period dtype directly.

2. DAX Measures (Power BI)

Active %

```
Active % =  
DIVIDE(  
    COUNTROWS(FILTER('cleaned_subscriptions',  
        'cleaned_subscriptions'[is_churned] = FALSE())),  
    COUNTROWS('cleaned_subscriptions')  
) * 100
```

Percentage of customers who have NOT churned.

Churn %

```
Churn % =  
DIVIDE(  
    COUNTROWS(FILTER('cleaned_subscriptions',  
        'cleaned_subscriptions'[is_churned] = TRUE())),  
    COUNTROWS('cleaned_subscriptions')  
) * 100
```

Percentage of customers who have cancelled. Active % + Churn % = 100.

Revenue Collected

```
Revenue Collected =  
CALCULATE(  
    SUM('cleaned_subscriptions'[subscription_cost]),  
    'cleaned_subscriptions'[was_subscription_paid] = "true"  
)
```

Total subscription cost for customers who actually paid. Check exact value of was_subscription_paid before using.

Churned Count

```
Churned Count =  
CALCULATE(  
    COUNTROWS('cleaned_subscriptions'),  
    'cleaned_subscriptions'[is_churned] = TRUE()  
)
```

Count of churned customers per time period. Used on the churn trend line chart.

Customer Count

```
Customer Count = COUNTROWS('cleaned_subscriptions')
```

Total number of customers. Used in the cohort matrix values field.

months_to_churn (Calculated Column)

```
months_to_churn =  
IF(  
    ISBLANK('cleaned_subscriptions'[canceled_date]),  
    BLANK(),  
    DATEDIFF(  
        'cleaned_subscriptions'[created_date],  
        'cleaned_subscriptions'[canceled_date],  
        MONTH  
    )  
)
```

Created as a Calculated Column (not a measure). Shows how many months after signup a customer cancelled. Blank for active customers.

3. Power BI Visuals

KPI Cards

Fields	Total Customers, Active %, Churn %, Revenue Collected
Notes	Format → Callout value → Display units → None (to show full numbers like 3,069 instead of 3K). Format to 2 decimal places.

Bar Chart — Churn by Cohort Month

Fields	X-axis: cohort_month Y-axis: Churn %
Notes	Originally intended as churn by subscription_interval but that column only contained 'month' so it was replaced with cohort_month.

Line Chart — Churn Trend Over Time

Fields	X-axis: canceled_date (Month level) Y-axis: Churned Count
Notes	Use Churned Count measure (not Churn %) to avoid 100% issue caused by filtering only cancelled rows. Shows monthly trend.

Matrix — Cohort Retention Heatmap

Fields	Rows: cohort_month Columns: months_to_churn Values: Customer Count
Notes	Apply conditional formatting: Format visual → Cell elements → Background color → fx → Field value → Customer Count.

Donut Chart — Paid vs Unpaid

Fields	Legend: was_subscription_paid Values: customer_id (Count)
Notes	Shows proportion of paid vs unpaid subscriptions across all customers.

Scatter Plot — Duration vs Cost

Fields	X-axis: subscription_duration_days (Don't summarize) Y-axis: subscription_cost (default) Legend: status Values: customer_id (Count)
Notes	Set X-axis (subscription_duration_days) to Don't summarize. Leave Y-axis (subscription_cost) as default. Add customer_id to Values.

Slicer — Cohort Month

Fields	Field: cohort_month
Notes	Added to fill blank dashboard space. Makes all visuals interactive — users can filter the entire dashboard by a specific cohort month.

4. Key Decisions & Lessons Learned

Why keep nulls in canceled_date?

Filling with a string like 'Not Cancelled' converts the column to object dtype, breaking DATEDIFF and duration calculations. Nulls are handled conditionally using IF(ISBLANK(...)) in DAX.

Why skip churn by subscription_interval?

The subscription_interval column only contained 'month' — no variety means no comparison. The visual was replaced with a cohort month bar chart which is more insightful.

Why use Churned Count instead of Churn % on the line chart?

Using Churn % on canceled_date as the X-axis filters to only cancelled rows, making every point 100%. Churned Count avoids this by counting rows without dividing by a filtered total.

months_to_churn as Calculated Column, not Measure

DAX measures aggregate across rows and throw a 'single value cannot be determined' error when used with row-level date functions like DATEDIFF. Calculated columns compute row by row, which is what we need here.

Why convert cohort_month to string before CSV export?

Python's Period dtype is not natively supported by Power BI's CSV reader. Converting to string (e.g., '2023-01') ensures it loads correctly and can be used as a text slicer or axis field.

Workflow: Notebook → CSV → Power BI

Cleaning in Python gives full control and reproducibility. Complex cohort logic is harder to do in Power Query. Exporting a clean CSV as a single source of truth keeps the Power BI file simple.